Bishnu Bhusal

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Summary

PhD candidate (Dec 2026) specializing in provable privacy, LLM safety, and formal verification, with a track record of building and shipping research systems validated at scale. Research internships at Los Alamos National Laboratory, Amazon, and SRI International. Published at top-tier venues including AAAI, ACM CCS, and OOPSLA. Brings 5+ years of software engineering experience delivering scalable ML pipelines and production systems.

Experience

Los Alamos National Laboratory March 2026 – Present
Research Intern *Los Alamos, NM*

- Designing and evaluating a **private LLM fine-tuning pipeline** for sensitive data; enforces formal (ε, δ)-DP guarantees while benchmarking privacy-utility trade-offs across multiple downstream tasks using standard NLP datasets and evaluation metrics.

SRI International Fall 2024
Research Intern *Menlo Park, CA*

- Built and evaluated a **differentially private ICL pipeline** that generates DP-safe synthetic demonstrations for LLM inference without fine-tuning; outperformed prior baselines on **5 NLP benchmarks** (text classification, QA) with up to **23% accuracy gains**; published as **first author at AAAI 2026**.

Amazon Web Services (AWS) Summer 2024
Applied Scientist Intern *Santa Clara, CA*

- Led design and prototyping of a **copyright protection system for RAG pipelines**, introducing an inference-time mechanism with **provable safeguards** against copyrighted content reproduction without requiring model fine-tuning.

University of Missouri Jan. 2022 – Present
Graduate Research Assistant *Columbia, MO*

- Built **DiPApprox**, the first automated system for verifying differential privacy in algorithms with continuous noise (Gaussian, Laplace) — a previously unsolved class of problems — enabling practical validation of real-world privacy mechanisms such as **Sparse Vector Technique** and **Noisy Max**.
- Developed a **DiP automata**-based verification system for differential privacy in online randomized algorithms, achieving up to **100× speedup** over CheckDP and enabling scalable analysis of complex multi-variable mechanisms.
- Extended dReal (**$\int$dReal**) in a cross-institution collaboration (UIUC, UIC, Univ. of Missouri), enabling **verification and synthesis** of privacy, fairness, and utility properties for probabilistic programs; achieved **2× speedup** over Mathematica on most benchmarks.
- Proposed a game-theoretic ransomware defense using strategic deception; won **Best Paper at GameSec 2024**. Developed the Arculus UAV communication system, adopted for training use by the **U.S. Marines**.

Geoedge 2020 – 2022
Software Engineer *Tel Aviv, Israel*

- Led migration of microservices handling **1.5B daily requests** to Kubernetes with custom auto-scaling, including a **multi-label advertisement image classifier**; cut cloud costs by **35%** and downtime by **25%** through improved CI/CD pipelines and deployment reliability.

Hashunited / Furitech / STS 2017 – 2020
Software Engineer *Tel Aviv, Israel*

- Built cryptocurrency trading and mining platforms for **70+ coins** with 99.9%+ uptime; developed exam management system (Java, Spring Boot) serving **1M+ students**, adopted by **20+ government agencies**.

Education

University of Missouri — Columbia, MO
Ph.D. in Computer Science — GPA 4.0 — Advisor: Prof. Rohit Chadha — *Expected July 2026*
Thesis: Automated Verification of Differentially Private Algorithms

University of Missouri — Columbia, MO
M.S. in Computer Science — GPA 4.0 — Minor: Statistics — Cert: Cybersecurity — *2024*

Tribhuvan University — Kathmandu, Nepal
B.Eng. in Computer Engineering — *2017*

Selected Publications

[1] **Bhusal B.** et al. "Privacy Preserving In-Context-Learning Framework for Large Language Models." *AAAI 2026*

[2] **Bhusal B.**, Chadha R., Sistla A.P., Viswanathan M. "Approximate Algorithms for Verifying Differential Privacy with Gaussian Distributions." *ACM CCS 2025*

[3] Rivera C., **Bhusal B.**, Chadha R., Sistla A.P., Viswanathan M. "Checking δ-Satisfiability of Reals with Integrals." *OOPSLA 2025*

[4] **Bhusal B.**, Ma Y., Chadha R. "Privacy Nutrition Labels: Promise, Practice, and Paradoxes in Communicating Privacy." *HCI International 2025*

[5] Neupane R.L., **Bhusal B.** et al. "On Countering Ransomware Attacks Using Strategic Deception." *GameSec 2024* — **Best Paper Award**

[6] Chadha R., Sistla A.P., Viswanathan M., **Bhusal B.** "Deciding Differential Privacy of Online Algorithms with Multiple Variables." *ACM CCS 2023*

Technical Skills

- **Privacy & Security**: Differential Privacy · Trustworthy AI · Formal Methods · Model Checking · SMT Solvers
- **Machine Learning**: Transformers · PyTorch · JAX · HuggingFace · Fine-Tuning · LoRA · RAG · ICL
- **Systems / Infra**: AWS (SAA-C03) · Azure · GCP · Kubernetes · Docker · Jenkins · CI/CD
- **Programming**: Python · C++ · RCoq · Lean · Java · JavaScript · SQL

Awards & Certifications

- **Best Paper Award** — GameSec 2024, Game-Theoretic Ransomware Defense
- **Winner, GPC Interdisciplinary Case Competition** — University of Missouri, 2024
- **Research Fellowship ($7,500)** — Quantum Innovation Center, Univ. of Missouri, 2025
- **Student Travel Grants** — ACM CCS 2025 · IEEE CSF 2024
- **EECS Excellence Fellowship** — University of Missouri, 2022
- **AWS Certified Solutions Architect – Associate (SAA-C03)**
- **Graduate Certificate in Cybersecurity** — University of Missouri
- **Deep Learning Specialization** — Coursera

Selected Talks

- **Private and Trustworthy AI with Provable Guarantees** — Invited Talk, University of Hawaii at Manoa, 2026
- **Automated Verification of Differentially Private Algorithms** — Invited Talk, AIST (National Institute of Advanced Industrial Science and Technology), Tokyo, 2025
- **Approximate Algorithms for Verifying Differential Privacy** — Conference Talk, ACM CCS 2025, Taipei
- **Deciding Differential Privacy of Online Algorithms** — Conference Talk, ACM CCS 2023, Copenhagen